4.3.4 Water – Cement Ratio

"Water-cement ratio" is defined as the ratio of the weight of mixing water to the weight of cement used in the concrete. The strength of concrete increases with decrease in water-cement ratio.

Water reacts with cement chemically and causes setting and hardening of concrete. It is found theoretically that water required is about 0.50 to 0.60 time the weight of cement.

4.3.5 Preparation of Cement Concrete

Cement concrete is prepared either by hand mixing or machine mixing depending on the requirement. The materials are mixed thoroughly, so that a uniform distribution of materials is obtained. The thorough mixing ensures that cement in the form of a film completely covers the surface of aggregate.

4.3.5.1 Hand Mixing

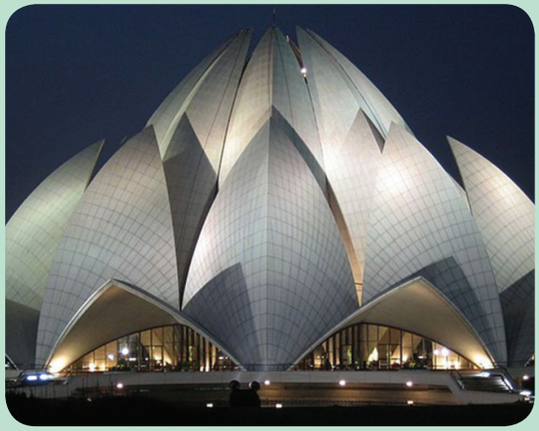


Mixing by hand is done either in a steel pan or on a pucca water tight platform. First, the sand and cement in the specified proportions are mixed thoroughly. Then, this mixture is spread evenly on a stack of coarse aggregate. It is turned over twice in dry state. Then the measured quantity of water is added and the mixture is mixed thoroughly to get a uniform mix. The prepared mix shall be consumed in 30 minutes after adding water. Hand mixing is best for small works.

DO YOU KNOW?

In India the buildings made out of white concrete is Bhai lotus temple (constructed in1986) located in Delhi.

Search link: http://en.m.wikipedia.org>wiki>lotustemple



4.3.5.2 Machine Mixing



Tilting Drum Concrete Mixer

Civil_English_Unit_4.indd 92 12/21/2019 2:47:49 PM